IoT-BASED SMART IRRIGATION AND SOIL MANAGEMENT SYSTEM

Final-Year Engineering Project Guide

ESP32 • Soil Moisture Monitoring • Automated Pump Control • IoT Dashboard

Prepared as a student implementation guide. Calibrate sensors and verify wiring before connecting a pump.

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1. Abstract and Problem Statement

This project uses an ESP32 to monitor soil moisture and control a low-voltage irrigation pump through a suitable relay or MOSFET driver. Optional DHT22 and tank-level sensors add environmental monitoring and dry-run protection. Wi-Fi can publish readings to ThingSpeak, while local threshold control continues if network access is lost.

Manual watering and fixed-time schedules may not reflect the actual moisture available around plant roots, which can cause overwatering, underwatering, unnecessary labour, and poor visibility into field conditions.

2. Objectives and Architecture

• Measure soil moisture near the root zone.

• Use configurable moisture thresholds with hysteresis.

• Stop the pump if the tank is empty, sensor configuration is invalid, or the maximum run time is reached.

• Publish readings and pump state to a remote dashboard when connected.

Architecture: soil moisture sensor (+ optional DHT22 and tank switch) → ESP32 controller → relay/driver → DC pump. The ESP32 also sends telemetry to ThingSpeak over Wi-Fi.

3. Hardware and Software

• ESP32 development board

• Capacitive soil-moisture sensor with analog output

• Properly rated relay module or MOSFET driver

• Low-voltage DC pump, compatible supply, tubing, wires and enclosure

• Optional DHT22 and float switch

• Arduino IDE or PlatformIO; ThingSpeak account for remote data

4. Working Principle

• Initialise sensors and set pump OFF.

• Read soil sensor and convert raw ADC values to a calibrated percentage.

• Start watering below the ON threshold.

• Stop watering at/above the OFF threshold.

• Apply tank-empty, invalid-reading, and maximum-run-time safety checks.

• Upload data periodically when Wi-Fi is available; retain local control offline.

5. Firmware Setup and Calibration

• Open firmware/smart_irrigation.ino in Arduino IDE and select the correct ESP32 board and port.

• Set Wi-Fi credentials and ThingSpeak Write API key locally; never publish real credentials.

• Measure ADC readings in dry and moist soil, then update MOISTURE_DRY_RAW and MOISTURE_WET_RAW.

• Set MOISTURE_ON_PERCENT and MOISTURE_OFF_PERCENT for your soil and crop. Keep ON lower than OFF.

• Verify relay polarity and test with the pump disconnected first.

• Use Serial Monitor at 115200 baud.

6. ThingSpeak Dashboard

Create a ThingSpeak channel and enable Field 1 (soil moisture), Field 2 (temperature), Field 3 (relative humidity), Field 4 (pump state), and Field 5 (tank available). Put the Write API key into the local firmware configuration. Temperature and humidity are sent only when USE_DHT is enabled and a DHT22 is connected.

7. Testing Plan

Test	Expected result
Dry soil + water available	Pump turns ON below threshold
Moist soil	Pump remains/stops OFF
Tank empty	Pump forced OFF
Wi-Fi unavailable	Local control continues
Maximum runtime reached	Pump stops
Cloud configured	Telemetry appears in channel

Record actual dry/wet ADC values, pump response time, water volume per cycle, cloud update interval, and observed faults. Do not report assumed measurements as experimental results.

8. Limitations and Future Enhancements

A single sensor represents only a small area. Calibration and sensor drift affect accuracy, and Wi-Fi/cloud services may be unavailable. The prototype does not itself prove water savings or crop-yield improvement. Future work can add multiple zones, rain/forecast data, solar power, secure authentication, and predictive recommendations.

9. Safety Notes

• Never drive a pump directly from an ESP32 GPIO.

• Use a driver and power supply rated for pump startup/current requirements.

• Keep mains wiring out of a student prototype unless supervised by a qualified person.

• Keep electronics dry and use appropriate enclosure/strain relief.

- Confirm relay polarity, sensor voltage, and wiring against component datasheets.